Exam info TARI29

TARI29 Artificial Intelligence, 7.5 credits. Exam Instructions

Allowed aids:

- VS Code and SWI-Prolog installed at the computers in the exam room
- SWI-Prolog documentation provided as PDF
- The logical equivalence sheet provided as PDF
- Other material provided as PDF
- Calculator

However, you are not allowed to use any other form of assistance. Specifically, it is of course *strictly prohibited* to collaborate with other students or use online resources during the exam.

Instructions for answering questions:

- Each question will provide a task description and the maximum amount of points that can be awarded for a correct answer.
- Type in your answers directly in the Inspera system.
- If a question *mentions* that you can submit a drawing or other type of writing on paper, you can get paper sheets from the personnel in the exam room. Then you need to hand in your sheets to the personnel. The paper sheets will be delivered to the examiners after the exam. If you write on paper, **write in legible (clear) handwriting** so that your answer is well readable.
- For *Prolog problems*, write and debug your code in the VS Code environment. Then copy the Prolog code of your solution from VS Code and paste it directly in the text field of the corresponding question.

Grading:

3	28–41.5 points
4	42–55.5 points
5	56–70 points

The distribution of points among question sections:

Section	Search	Game Theory	MDP	Logic	Prolog
Points	12	12	9	14	23

Good Luck!

1 Uninformed Search

Your responses in text fields are saved automatically

Uninformed Search

We have studied the 8-puzzle in the course when discussing search algorithms. Below are four statements about solving the 8-puzzle with uninformed search. For each statement, state whether it is true or false and provide a short motivation for your answer. Each statement is worth 4 points.

- Iterative deepening search will always find a solution.
- Using breadth-first search, the first solution found will always be optimal, i.e., have the minimal number of moves.
- Using depth-first search, the first solution found will always be optimal, i.e., have the minimal number of moves.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 12

2 game-theory-task1_3

Game Theory - Task 1

Consider the payoff matrix below for a game in normal form.

		Player B		
		X	Y	Z
Player A	X	2,0	1,1	4,2
	Y	3,4	1,2	2,3
	Z	1,3	0,2	3,0

Are there any pure Nash Equilibria or Pareto Optimal outcomes in the game? If so, what are they (and why)?

You may use paper sheets to answer this question.
Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 5

3 game-theory-task2_Ericsson

Game Theory - Task 2

Let us assume the following scenario. Ericsson, a Swedish multinational networking and telecommunications company, is working on a new chip that leverages Quantum Computing. At the same time, IBM is working on a similar chip. Due to the ambitious type of this task, they reached out to a research laboratory run by an established professor at Université de Sherbrooke, Canada. Ericsson and IBM can either team up and work on a collaborative project with the professor's group, or each can work independently with the group to develop their own chip. Based on their decision, the following cases may occur:

- if they team up and agree to work on a collaborative project with the professor's group, then the project costs will be divided, and the collaboration between companies will speed up the project. The expected profits in that case are 3 billion SEK for each company.
- if they defect and work on separate projects with the professor's group, they will have increased costs, and additional resources will be required since the research laboratory will be focusing on two different ideas. Eventually, the expected profits, in that case, will equal 1 billion SEK for each company.
- but if one cooperates (i.e., starts working on an agreed common project) and the other defects (i.e., even though they have agreed on a common project, this company starts working on a side-project using information from the common project), then the company who defected will use the technology faster and thus, will earn a profit of 4 billion SEK, while the cooperating company will have no profits.

Represent the above situation as a game in its Normal Form. What is the dominant strategy of each company and why? Can we predict what would be the outcome of this game? Do we make any assumptions to make this prediction?

You may use paper sheets to answer this question.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 3

4 game-theory-task3_4

Game Theory - Task 3

Calculate a Nash Equilibrium with mixed strategies from the payoff matrix below and state the value of the game at this equilibrium. Use the method of assuming the opponent should be indifferent between its actions. Show all your calculations.

		Player B	
		X	Y
Player A	X	2,-3	1,2
	Y	1,1	4,-1

You may use paper sheets to answer this question.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 4

5 MDP-task1_3

MDP / Reinforcement Learning - Task 1

Assume an MDP consists of the two states S_1 and S_2, the two actions A_1 and A_2, a *discount factor* $\gamma = 1/2$, a *reward function* R and a *transition function* T as shown below:

$$R(s,a,s') = \begin{cases} 1/2 & \text{if } s' = S_1 \\ -1/2 & \text{if } s' = S_2 \end{cases}$$

s	a	s'	T(s,a,s')
S_1	A_1	S_1	$3/4$
S_1	A_1	S_2	$1/4$
S_1	A_2	S_1	$1/2$
S_1	A_2	S_2	$1/2$
S_2	A_1	S_1	$1/3$
S_2	A_1	S_2	$2/3$
S_2	A_2	S_1	$1/2$
S_2	A_2	S_2	$1/2$

Perform one iteration of *value iteration* and enter the *value* for $V_1(S_1)$ in the table below (you don't have to enter a value into the gray cell). Use the initial values for $V_0(s)$ below instead of initializing all values to 0. Show all your calculations.

s	$V_0(s)$	$V_1(s)$
S_1	4	?
S_2	2	

$$V_{k+1}(s) \leftarrow \max_a \sum_{s'} T(s,a,s')[R(s,a,s') + \gamma V_k(s')]$$

You may use paper sheets to answer this question.
Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 3

6 MDP-task2_2024_2

MDP / Reinforcement Learning - Task 2

Assume an MDP consists of the two states S_1 and S_2, the two actions A_1 and A_2.

Assume *Q-learning* with a *learning* rate $\alpha = 1/4$ and without discount (i.e., $\gamma = 1$) is used. The following episode is observed (i.e. first the agent executes the action A_2 in state S_2, ending up in state S_2 with the reward *-1*, followed by the agent executing the action A_1 in state S_2, ending up in state S_1 with the reward *1*):

s_1	a_1	r_1	s_2	a_2	r_2	s_3
S_2	A_2	-1	S_2	A_1	1	S_1

Use the initial *Q-values* Q_0 in the table below, and enter the values for Q_1 and Q_2. Show all your calculations.

s	a	$Q_0(s,a)$	$Q_1(s,a)$	$Q_2(s,a)$
S_1	A_1	$1/2$	?	?
S_1	A_2	$1/8$	?	?
S_2	A_1	$5/4$	?	?
S_2	A_2	$1/4$	?	?

$$Q_{k+1}(s,a) \leftarrow Q_k(s,a) + \alpha[r + \gamma \max_{a'} Q_k(s',a') - Q_k(s,a)]$$

You may use paper sheets to answer this question.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 6

7 logic-fol-wizard-taks1

Logic: First-order logic—Task 1

Consider the following knowledge base, given in natural language:

```
Harry is not evil.
Every wizard knows Harry.
There is an evil wizard who smokes.
Every wizard who knows Harry also knows Dumbledore.
```

Translate the knowledge base into first-order logic, using one FOL sentence for each sentence above.

You must use the following predicates and constants:

- *Unary predicates*; i.e. taking one argument: wizard, smokes, evil.
- *Binary predicates*; i.e. taking two arguments: knows.
- *Constants*; i.e. names of people: Harry, Dumbledore

You may use paper sheets to answer this question.

Alternatively, you can use the following way to quickly type in logical formulae in Inspera.

- For logical connectives, use symbols that are used in the programming languages Java, C\#, and C++: '!' for negation, '&&' for conjunction, '||' for disjunction, '->' for implication , '' <--> for biconditional
- For universal quantifier use 'forall', and for existential quantifier use 'exists'.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 4

8 logic-fol-wizard-task2

Logic: First-order logic—Task 2

Consider the following knowledge base consisting of one first-order logic sentence:

$$\forall xy.\, (\mathbf{wizard}(x) \wedge \mathbf{wizard}(y) \Rightarrow \mathbf{knows}(x, y))$$

Define two first-order models, one for which this sentence is true and one for which this sentence is false. Formally specify the domain and an interpretation function relative to this domain for each model.

You may use paper sheets to answer this question.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 4

9 logic-fol-wizard-task3

Logic: First-order logic—Task 3

Consider the following knowledge base consisting of one first-order logic sentence:

$$\forall xy.\, (\mathbf{wizard}(x) \wedge \mathbf{wizard}(y) \Rightarrow \mathbf{knows}(x, y))$$

Use resolution to proof that this knowledge base entails the query sentence

$$\mathbf{wizard}(\mathbf{Dumbledore}) \Rightarrow \mathbf{knows}(\mathbf{Dumbledore}, \mathbf{Dumbledore})$$

This proof requires you to convert the knowledge base and the negated query into a set of clauses. Recall that the steps for doing this are the following:

1. Eliminate biconditionals and implications;
2. Move negation inwards, using the logical equivalences;
3. Standardize variables if necessary: each quantifier should use a different variable;
4. Drop existential quantifiers and replace existential variables by Skolem functions of the enclosing universally quantified variables;
5. Drop universal quantifiers;
6. Use distributivity law to distribute disjunction over conjunction.

You may use paper sheets to answer this question.
Alternatively, you can use the following way to quickly type in logical formulae in Inspera.

- For logical connectives, use symbols that are used in the programming languages Java, C\#, and C++: '!' for negation, '&&' for conjunction, '||' for disjunction, '->' for implication , '' for biconditional
- For universal quantifier use 'forall', and for existential quantifier use 'exists'.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 6

10 Prolog-CLP-auburn

Prolog: Constraint Logic Programming (CLP)

Define the predicate **puzzle(Solution)** that solves the cryptarithmetic puzzle below using CLP(FD). You need to find all concrete solutions.

```
R U B Y
+ P E A R
+ B R A S S
-------------
 A U B U R N
```

Use separate lists to represent each word, so that after running the query **?-puzzle(Solution)**, each word in a solution of the puzzle would be represented by a list of digits. Like in the example given in the lecture:

```
Solution = ([9, 5, 6, 7] +[1, 0, 8, 5]=[1, 0, 6, 5, 2])).
```

Copy the Prolog code of your solution from VS Code and paste it directly in this text field.

Maximum marks: 8

11 Prolog-list-ordered-desc

Prolog: Lists

Define the predicate **ordered_desc(List)** which is true if **List** is a list of numbers ordered in the descendent manner. One-element lists are considered to be ordered.

For example, this query

```
?- ordered_desc([12,9,6,5,5,3]).
```

would evaluate to *true*.

You have to use *cut* ('!') in your definition. You may <u>not</u> use any built-ins (library predicates are <u>not</u> allowed) in this task.

Copy the Prolog code of your solution from VS Code and paste it directly in this text field.

Maximum marks: 7

12 Prolog-data-postorder_tree

Prolog: Data structures

Write a procedure **postorder_tree/2** that collects all the nodes in a binary tree using post-order traversal: left subtree, right subtree, and root. Use the atom **nil** to represent an empty tree, and the term **t(L,X,R)** to represent a non-empty binary tree, where **X** denotes the root node and **L** and **R** denote the left and right subtree, respectively. For example:

```
?- postorder_tree( t( t(nil, b, nil), a, t( t(nil, d, nil), c, nil ) ), L).
L = [b, d, c, a].
```

Hint: Use a predicate for list concatenation from the SWI-Prolog library of predicates.

Copy the Prolog code of your solution from VS Code and paste it directly in this text field.

Maximum marks: 8